

BLVNK Tech Solutions — Municipal AI Dispatch Engine

Comprehensive Technical Documentation & API Architecture Guide

System Version: 2.0.0 (Next.js 15 App Router)	Target OS: Cross-Platform (Linux/Vercel/Windows)
Production Live URL: https://municipal-dispatch-ai-engine.vercel.app	AI Engine: Groq Qwen 2.5 (qwen-2.5-32b)
GitHub Repo: MilaniKubuli/Buildathon-Blvnk-Solution	Database: Supabase / Neon PostgreSQL

1. Executive Summary

The **BLVNK Municipal AI Dispatch Engine** is an enterprise-grade civic technology platform built for South African municipal dispatch centers and citizens. It streamlines service request intake, automatically detects and translates all **11 official South African languages**, performs zero-hallucination duplicate analysis using **Groq Qwen 2.5 AI**, pinpoints location data on interactive Leaflet maps, and synchronizes real-time tickets with **Supabase / Neon PostgreSQL**.

2. API Specification & Endpoints

2.1 AI Reasoning Endpoint: POST /api/analyze-report

Processes raw report text in any South African language, compares it against existing active incidents, assesses urgency, detects duplicates, and extracts structured location data.

```
POST /api/analyze-report HTTP/1.1
Host: municipal-dispatch-ai-engine.vercel.app
Content-Type: application/json

{
  "newReport": "Amanzi ayaphuma emgwaqweni e-Jan Smuts Ave Rosebank. Heavy clean water bursting onto sidewalk.",
  "existingIncidents": [{ "id": "INC-101", "category": "water_leak", "location": "Jan Smuts Ave", "lat": -26.1465, "lng": 28.0436 }],
  "languageCode": "zu"
}

HTTP/1.1 200 OK
Content-Type: application/json

{
  "success": true,
  "data": {
    "category": "water_leak",
    "location": "Jan Smuts Ave & 7th Ave",
    "suburb": "Rosebank",
    "landmark": "Near Shell Garage / Rosebank Mall",
    "urgency": "high",
    "short_summary": "Clean water pipe burst flooding pavement on Jan Smuts Ave.",
    "english_translation": "Water is coming out on the street in Jan Smuts Ave Rosebank near the garage.",
    "possible_duplicate": true,
    "matched_incident_ids": ["INC-101"],
    "detected_language": { "code": "zu", "name": "isiZulu", "confidence": 0.98 },
    "ai_provider": "Groq Qwen AI",
  }
}
```

```
    "ai_model": "qwen-2.5-32b"
  },
  "processingTimeMs": 82
}
```

2.2 System Health Check Endpoint: GET /api/health

```
GET /api/health HTTP/1.1
Host: municipal-dispatch-ai-engine.vercel.app

HTTP/1.1 200 OK
{
  "status": "ok",
  "framework": "Next.js 15 App Router",
  "hasGroqKey": true,
  "groqModel": "qwen-2.5-32b",
  "timestamp": "2026-08-01T13:28:22.917Z"
}
```

3. Multi-Lingual Configuration (11 Official SA Languages)

The system features comprehensive localized dictionaries and dynamic AI translation models for all 11 official South African languages:

Language Code	Language Name	Primary Native Regions	Sample Translation Key
en	English	Nationwide	Incident Log
zu	isiZulu	KwaZulu-Natal, Gauteng	Ilogi Yezehlakalo
xh	isiXhosa	Eastern Cape, Western Cape	Iskripthi Sezehlakalo
af	Afrikaans	Western Cape, Northern Cape	Insidenterog
nso	Sepedi (Northern Sotho)	Limpopo, Mpumalanga	Loga ya Ditiragalo
tn	Setswana	North West, Northern Cape	Khutso ya Ditiragalo
st	Sesotho (Southern Sotho)	Free State, Gauteng	Loga ya Ketsahalo
ts	Xitsonga	Limpopo, Mpumalanga	Rhekhodo ya Swiendlakalo
ss	siSwati	Mpumalanga	Ilogi Yetehlakalo
ve	Tshivenda	Limpopo	Mgodo wa Zwiitelea
nr	isiNdebele	Mpumalanga, Gauteng	Ilogi Yezehlakalo

4. AI Model & Reasoning Engine

The platform uses Groq's high-speed inference hardware running **Qwen 2.5 (qwen-2.5-32b)**, achieving sub-100ms response times. The engine categorizes municipal issues into: *water_leak*, *electricity_outage*, *pothole_traffic*, *illegal_dumping*, *sewage_overflow*, *fallen_tree*, and *missing_manhole*.

5. Logic Variables & Schema Definitions

```
export interface Incident {
  id: string; // Format: 'INC-101'
  title: string; // Headline
  category: IncidentCategory; // Municipal category key
  location: string; // Address / intersection
  suburb?: string; // Suburb / Ward
  landmark?: string; // Recognizable landmark
  urgency: 'low' | 'medium' | 'high';
  description: string; // Full citizen report
  reportedAt: string; // ISO Timestamp
  status: 'open' | 'in_progress' | 'resolved';
  reportedBy: string; // Citizen handle
  votes: number; // Upvote count
  lat: number; // Latitude (e.g. -26.2041)
  lng: number; // Longitude (e.g. 28.0473)
  languageCode?: SALanguageCode; // Language identifier (en, zu, xh...)
}
```

6. Full-Scale Application Integration

Landing Page Layout: Set to default tab 'database' (Incident Log) in src/App.tsx.

Questionnaire Wizard: 5-step guided intake UI for structured data collection.

Geospatial Centering: Interactive Leaflet map centered at South Africa (Johannesburg/Pretoria/Cape Town/Durban).

Database Sync: Real-time sync with Supabase and Neon PostgreSQL via NEON_CONNECTION_STRING.